



Probability

→ Random Experiment

An experiment / observation which maybe repeated a large no of times under very nearly ideal conditions and possible outcome of any particular observation is unpredictable but all possible outcomes can be described prior to its performance is called random experiment. eg - tossing a coin or rolling a die.

→ Sample points / event points

The outcomes of random experiment are called sample points.

→ Sample space (S)

It's the set of all sample points.

→ Event

It's any subset of sample space. eg → throwing 2 coins, $A = \{TH, HT\}$ is event because $A \subseteq S$

→ Certain event

An event having all sample points. It will definitely occur.

→ Impossible event (ϕ)

An event which has no sample points



→ Complementary event ($\bar{A}$)

For any event A , there is a set containing sample points which aren't in A . This event is called complementary event of A .

→ Simultaneous occurrence of 2 events (A_1, A_2)

Let A_1, A_2 be 2 events. Then set $A_1 \cap A_2$ represents simultaneous occurrence of A_1 & A_2 .

→ At least one of 2 events ($A_1 + A_2$)

Let A_1, A_2 be 2 events, then set $A_1 \cup A_2$ represents at least one of them occur.

→ Disjoint / Mutually exclusive events

If 2 events A_1 & A_2 have no common sample points $A_1 \cap A_2 = \emptyset$ then A_1 & A_2 are mutually exclusive.

→ Pairwise disjoint events

Let $A_1, A_2, \dots, A_n$ be n events. Events A_i ($i=1, 2, \dots, n$) are said to be disjoint pairwise if no two of them have any common sample points.

$$A_i \cap A_j = \emptyset \quad i \neq j \text{ and } i, j = 1, 2, \dots, n$$

→ Exhaustive events

Two or more events are said to be exhaustive if at least one of them necessarily occurs or in other words the events $A_1, A_2, \dots, A_n$ are exhaustive if $A_1 \cup A_2 \cup A_3 \dots \cup A_n = S$.

→ Elementary / Simple event

An event containing exactly one sample point
eg - $A_1 = \{1\}$



- **composite event**
An event which can be decomposed into simple events i.e. which can be expressed as union of two or more simple points
- **Equally likely sample points**
The sample points of a sample space are said to be equally likely if none of them may not be expected rather than other
- **classical definition of probability**
Let's suppose that random experiment E is such that its sample space S contains finite no. of events $n(S)$ of sample points all of which are equally likely. Then probability of event A which contains $n(A)$ sample points is defined by

$$P(A) = \frac{n(A)}{n(S)}$$

Q Probability of leap year having 53 Mondays

$$7 \sqrt{366} 152$$

$$\begin{array}{r} -35 \\ \hline 16 \end{array}$$

$$\begin{array}{r} 16 \\ -14 \\ \hline 2 \end{array}$$

$$\begin{array}{r} 2 \\ -14 \\ \hline 2 \end{array}$$

$$2$$

Total $\rightarrow 7$

Fav $\rightarrow 2$

$$P(53 \text{ mon}) = \frac{2}{7}$$

These 2 days can be

SM or MT

Q A card is drawn from deck of 52 cards. Find probability that it's

i) an ace

4 suits $\rightarrow$ 4 ace

$$P(\text{ace}) = \frac{4}{52} = \frac{1}{13}$$

ii) a heart

1 suit $\rightarrow$ 13 heart cards

$$P(\text{heart}) = \frac{13}{52} = \frac{1}{4}$$

iii) a 9 or club

13 club + 3 9's from H, S, D

$$= \frac{16}{52} = \frac{4}{13}$$

iv) neither spade nor 10

$$= \frac{16}{52} = \frac{4}{13}$$

$\rightarrow$ Axiomatic Definition of Probability

Let E be a random experiment & S be its sample space. Let P be a function from class of all events to set of all real nos satisfying following axioms :-

i) $P(A) \geq 0$

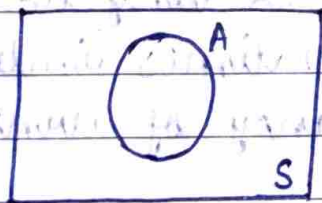
ii) $P(S) = 1$

iii) If $A_1, A_2, \dots$ be a finite/infinite sequence of pairwise mutually exclusive events then

$$P(A_1 \cup A_2 \cup \dots) = P(A_1) + P(A_2) + \dots$$

Then for any event A , $P(A)$ denotes probability of an event A

Q P.T. $P(\bar{A}) = 1 - P(A)$ $\bar{A} \rightarrow$ complimentary event of A



since, $A \cup \bar{A} = S$

also $A \cap \bar{A} = \phi$

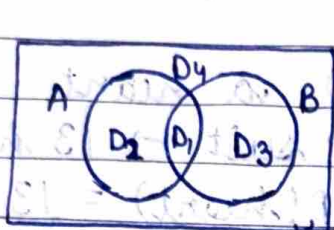
$$P(A \cup \bar{A}) = P(S) = 1$$

$$P(A) + P(\bar{A}) = 1$$

$$\Rightarrow P(\bar{A}) = 1 - P(A)$$



Q. P.T. $P(\bar{A} \cap \bar{B}) = P(\bar{A}) - P(B) + P(AB)$



$$D_1 = A \cap B$$

$$D_2 = A \cap B^c = A \cap \bar{B}$$

$$D_3 = B \cap A^c = B \cap \bar{A}$$

$$D_4 = A^c \cap B^c = \bar{A} \cap \bar{B}$$

To prove $P(\bar{A} \cap \bar{B}) = P(\bar{A}) - P(B) + P(AB)$
 $= P(\bar{A} \cap \bar{B}) = P(\bar{A}) - P(B) + P(A \cap B)$

LHS :-

$$P(\bar{A} \cap \bar{B}) = P(D_4)$$

RHS :-

$$P(\bar{A}) - P(B) + P(A \cap B)$$

$$P(D_2 \cup D_4) - P(D_1 \cup D_3) + P(D_1)$$

Now D_1, D_3, D_4 are pairwise exclusive

$$\Rightarrow P(D_3 \cup D_4) = P(D_3) + P(D_4)$$

$$\Rightarrow P(D_1 \cup D_3) = P(D_1) + P(D_3) \quad \left. \begin{array}{l} \Rightarrow P(D_3 \cup D_4) = P(D_3) + P(D_4) \\ \Rightarrow P(D_1 \cup D_3) = P(D_1) + P(D_3) \end{array} \right\} \text{axiom-iii}$$

RHS :-

$$\Rightarrow P(D_3) + P(D_4) - P(D_1) - P(D_3) + P(D_1)$$

$$= P(D_4)$$

$$\Rightarrow \text{LHS} = \text{RHS}$$

Hence Proved

→ Frequency definition of probability

Suppose we have random experiment E with sample space S . Let $A \subseteq S$ be any event. Let us repeat random exp. n times independently. Let X_n is relative frequency of events during these trials.

Then it's seen that as $n \rightarrow \infty$

the random variable X approaches some fixed number. This behaviour is an example of statistical regularity. Limiting fixed number is called probability of event. This is frequency definition of probability.

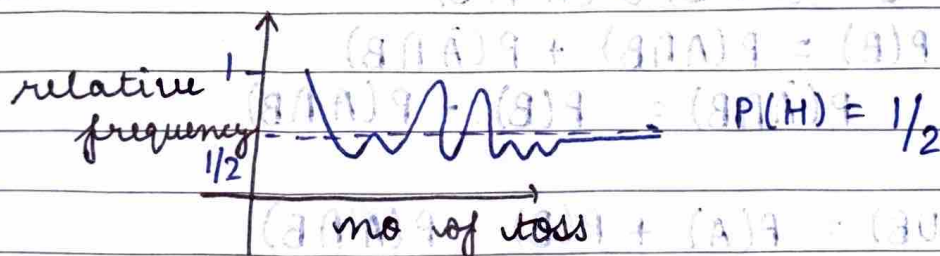
$$P(A) = \lim_{n \rightarrow \infty} \frac{f(A)}{n}$$

eg $\rightarrow$ tossing a coin

$E \rightarrow$ Tossing a coin

$A \rightarrow$ getting Head

No of toss	1	2	3	4	5
outcome	H	T	H	H	T
Frequency	1	1	2	3	3
Relative freq		1/2	2/3	3/4	3/5
= Freq / No of toss					



$\rightarrow$ Theorems in probability

1. $0 \leq P(A) \leq 1$ for any event A .

Let E be a random experiment that's repeated in times under identical conditions. Let A be an event which occurs $n(A)$ times. Then

$$0 \leq n(A) \leq n$$

$$\Rightarrow 0 \leq \frac{n(A)}{n} \leq 1$$

$$\Rightarrow 0 \leq P(A) \leq 1$$



2. $P(S) = 1$ and $P(\phi) = 0$: $S \rightarrow$ certain event
 $\phi \rightarrow$ impossible event

$$P(S) = \frac{n(S)}{n} = \frac{n}{n} = 1$$

$$P(\phi) = \frac{n(\phi)}{n} = \frac{0}{n} = 0$$

3. If the events $A_1, A_2, \dots, A_n$ are mutually exclusive then
- $$P(A_1 \cup A_2 \cup \dots \cup A_n) = P(A_1) + P(A_2) + \dots + P(A_n)$$

4. $\star$ Total Probability

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B) = P(A \cup (\bar{A} \cap B))$$

$$= P(A) + P(\bar{A} \cap B) \quad \because (\bar{A} \cap B) \text{ and } A$$

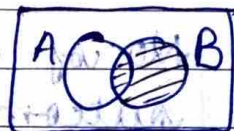
Again

$$B = (A \cap B) \cup (\bar{A} \cap B)$$

$$\Rightarrow P(B) = P(A \cap B) + P(\bar{A} \cap B)$$

$$\therefore P(\bar{A} \cap B) = P(B) - P(A \cap B)$$

$$\Rightarrow P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



5. If A_1, A_2, A_3 are any three events not necessarily mutually exclusive then

$$P(A_1 \cup A_2 \cup A_3) = P(A_1) + P(A_2) + P(A_3) - P(A_1 \cap A_2)$$

$$- P(A_2 \cap A_3) - P(A_1 \cap A_3) + P(A_1 \cap A_2 \cap A_3)$$

6. $P(\bar{A}) = 1 - P(A)$

7. If $A_1, A_2, \dots, A_n$ are mutually exclusive and exhaustive events then

$$\sum_{i=1}^n P(A_i) = 1$$

8. For any two events A_1 & A_2 where $A_1 \subset A_2$

i) $P(A_1) \leq P(A_2)$

ii) $P(A_2 - A_1) = P(A_2) - P(A_1)$

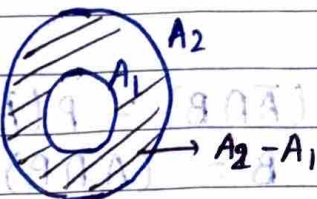
i) $A_1 \subset A_2$

$$\Rightarrow n(A_1) \leq n(A_2)$$

$$\Rightarrow \frac{n(A_1)}{n} \leq \frac{n(A_2)}{n}$$

$$\Rightarrow P(A_1) \leq P(A_2)$$

$$\Rightarrow P(A_1) \leq P(A_2)$$



ii) $A_2 = A_1 \cup (A_2 - A_1)$

$$P(A_2) = P(A_1) + P(A_2 - A_1)$$

$$P(A_2 - A_1) = P(A_2) - P(A_1)$$

A ke hone ki probab
Tab B already ho chuka
hai

→ Independent events

If for two events A & B , $P(A/B) = P(A)$ i.e. chance of occurrence of event A isn't affected by occurrence of event B then A & B are independent events

$$P(A \cap B) = P(A) \cdot P(B)$$

only when A & B are independent

Q If A & B are independent then show following pairs are also independent

i) $\bar{A}$ & $\bar{B}$

ii) A & $\bar{B}$

iii) $\bar{A}$ & B

i) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$P(A \cup B) = P(A) + P(B) - P(A)P(B)$$

$$P(A \cup B) = P(\bar{A}) + P(B) (1 - P(A)) = P(\bar{A}) + P(B)P(\bar{A})$$

Taking complement

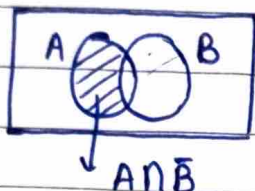
$$P(\bar{A} \cap \bar{B}) = P(\bar{A}) + P(\bar{B})P(A) = 1 - P(A) + P(A)P(\bar{B})$$

$$P(\bar{A} \cap \bar{B}) = 1 - P(A) (1 - P(\bar{B})) = 1 - P(A)P(B)$$

$$\Rightarrow P(\bar{A} \cap \bar{B}) = P(\bar{A})P(\bar{B})$$

ii) $P(A \cap \bar{B}) = P(A)P(\bar{B}) \rightarrow$ To Prove

LHS $\rightarrow P(A \cap \bar{B})$



$$\begin{aligned} A &= (A \cap B) \cup (A \cap \bar{B}) \\ \Rightarrow P(A) &= P(A \cap B) + P(A \cap \bar{B}) \\ \Rightarrow P(A \cap \bar{B}) &= P(A) - P(A \cap B) \\ &= P(A)(1 - P(B)) \\ &= P(A)P(\bar{B}) \end{aligned}$$

iii) $P(\bar{A} \cap B) = P(\bar{A})P(B)$

$$B = (A \cap B) \cup (\bar{A} \cap B)$$

$$P(B) = P(A \cap B) + P(\bar{A} \cap B)$$

$$\Rightarrow P(\bar{A} \cap B) = P(B) - P(A \cap B)$$

Q. A speaks truth in 75% cases while B speaks truth in 80% cases. What's P that they contradict each other?

$$P(E) = P(\bar{A})P(B) + P(\bar{B})P(A)$$

$$= \left(\frac{25}{100} \times \frac{80}{100} \right) + \left(\frac{20}{100} \times \frac{75}{100} \right)$$

$$= \left(\frac{1}{4} \times \frac{4}{5} \right) + \left(\frac{1}{5} \times \frac{3}{4} \right)$$

$$= \frac{7}{20} = 0.35$$

$\rightarrow$ Conditional Probability

We consider 2 events A & B connected with a random experiment E. Let's assume A occurs

$n(A)$ times and B occurs simultaneously with A $n(A \cap B)$ times for n repetitions. Then ratio $\frac{n(A \cap B)}{n(A)}$ is called conditional probability of B on the hypothesis that A has already occurred & is

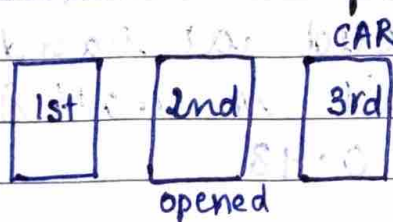
denoted by $P(B/A)$

$$P(B/A) = \frac{n(A \cap B)}{n(A)} = \frac{n(A \cap B)}{n} \times \frac{n}{n(A)} = \frac{P(A \cap B)}{P(A)}$$

$$P(B/A) = \frac{P(A \cap B)}{P(A)} \quad \text{where } P(A) \neq 0$$

$$\text{Hence } P(A/B) = \frac{P(A \cap B)}{P(B)}$$

* In a reality show, there are 3 doors, closed. Behind only 1 door there is a brand new car and behind other 2 is nothing. Host calls a person randomly to select a door. Suppose person has selected 1st door but car is behind 3rd door. The host opens second door & asks person whether he wants to change his answer. So ques is should he change his answer? However host will never open door behind which car is at the very first place whether the person has selected correct door.



If he doesn't switch, $P = 1/3$

solve using Bayes' theorem

$A \rightarrow$ car is behind door 1

$B \rightarrow$ car is behind door 2

$C \rightarrow$ car is behind door 3

$E \rightarrow$ door 2 is opened

$$P\left(\frac{C}{E}\right) = \frac{P(C) \times P(E/C)}{P(E)}$$

$\rightarrow$ If he changes his choice

$$P(C) = 1/3 \quad P(E/C) = 1 \quad \left[\begin{array}{l} \text{car is behind door 3 \& we chose door 1 so} \\ \text{host will open door 2} \end{array} \right]$$

$$P(E) = 1/2 \quad [\text{Host never open door 3}]$$

$$P(C/E) = \frac{1/3}{1/2} = \frac{2}{3} = 0.66$$

$2/3 > 1/3$
so he switch



Q $P(A) = 1/2$ $P(B) = 1/3$ $P(A \cap B) = 1/4$
 $P(A/B) = ?$

$$= \frac{P(A \cap B)}{P(B)} = \frac{1/4}{1/3} = \frac{3}{4}$$

Q $P(A \cap B \cap C) = 0$ P.T. $P(X/C) = P(A/C) + P(B/C)$
 where $X = A + B$
 $\Rightarrow X = A \cup B$

Now LHS

$$\begin{aligned} P(X/C) &= \frac{P(X \cap C)}{P(C)} = \frac{P((A \cup B) \cap C)}{P(C)} \\ &= \frac{P((A \cap C) \cup (B \cap C))}{P(C)} = \frac{P(A \cap C) + P(B \cap C)}{P(C)} \\ &= \frac{P(A \cap C)}{P(C)} + \frac{P(B \cap C)}{P(C)} = P(A/C) + P(B/C) \end{aligned}$$

Q In exam, 30% fail phy, 25% math & 12% fail both. A student is selected at random

i) $P(\text{fail in phy, known fail in maths})$

$$= \frac{P(P \cap M)}{P(M)} = \frac{12}{25} = 0.48$$

ii) $P(\text{fail at least one of them}) = P(A \cup B) = P(P \cup M)$

$$= P(P) + P(M) - P(P \cap M)$$

$$= \frac{30 + 25 - 12}{100} = 0.43$$

iii) $P(\text{pass at least one})$

$$P(\bar{P} \cup \bar{M}) = 1 - P(P \cap M)$$

$$= 1 - 0.12$$

$$= 0.88$$



Q A & B are independent events' witnesses. The $P(A \text{ speaks truth})$ is x while $P(B \text{ says truth})$ is y . A & B agree in certain statement. P.T.
 $P(\text{statement is true}) = \frac{xy}{1-x-y+2xy}$

Both agree $\rightarrow$ Both truth / Both false

Both true $\rightarrow x * y = xy$

Both false $\rightarrow (1-x)(1-y) = 1-x-y+xy$

$\Rightarrow$ Both agree $\rightarrow 1-x-y+2xy$ cases

$$P(\text{Stat is true}) = \frac{P(\text{Both true})}{P(\text{Both agree})} = \frac{xy}{1-x-y+2xy}$$

$\rightarrow$ Bayes Theorem

$$P(E_i/A) = \frac{P(E_i) P(A/E_i)}{P(E_1)P(A/E_1) + P(E_2)P(A/E_2) + P(E_3)P(A/E_3)}$$

Q An urn has 5 Red & 5 black balls. A ball is drawn at random, colour is noted & it's returned to urn. Moreover 2 additional balls of colour drawn are put into urn then ball is drawn at random. $P(\text{second ball is red}) = ?$

A $\rightarrow$ Red Ball is drawn first

B $\rightarrow$ Black Ball is drawn first

E $\rightarrow$ second ball is red

$$P(E) = P(A)P(E/A) + P(B)P(E/B)$$

$$= \frac{5}{10} \times \frac{7}{12} + \frac{5}{10} \times \frac{5}{12} = \frac{1}{2} \left[\frac{12}{12} \right] = \frac{1}{2}$$

Q. A manufacturer has 3 machines A, B, C. A produce 1% defective items while B → 5% & C → 7%. A is on job for 50% of time B is on job 30% time & C is for 20% time. A defective item was produced. $P(\text{A produce defective item}) = ?$

- $A_1 \rightarrow$ A was on work
- $A_2 \rightarrow$ B was on work
- $A_3 \rightarrow$ C was on work
- $E \rightarrow$ defective item produced

$$P(E|A) = \frac{P(A) \times P(A|E)}{P(A)P(A|E) + P(A_2)P(A_2|E) + P(A_3)P(A_3|E)}$$

$$= \frac{50}{100} \times \frac{1}{100}$$

$$\frac{50}{100} \times \frac{1}{100} + \frac{30}{100} \times \frac{5}{100} + \frac{20}{100} \times \frac{7}{100}$$

$$= \frac{5 \times 10^{-3}}{(5 + 15 + 14) \times 10^{-3}}$$

$$= \frac{5}{34}$$

$$P(E) = P(A)P(E|A) + P(B)P(E|B) + P(C)P(E|C)$$

$$= \left(\frac{50}{100}\right) \left(\frac{1}{100}\right) + \left(\frac{30}{100}\right) \left(\frac{5}{100}\right) + \left(\frac{20}{100}\right) \left(\frac{7}{100}\right)$$



→ Random variable

Let S be a given sample space. Then a real valued func X defined on S is called a random or stochastic variable or sometimes a variate. Thus for every point of sample space S we have a unique real value of X .

Range of X is set of all values assumed by X & is called spectrum of random variable.

• Discrete Random variable

Random variable X is said to be discrete if the spectrum of X is finite / countably ∞ i.e. an ∞ sequence of distinct values

• Continuous Random variable

X is said to be continuous if it can assume every value in an interval.

PMF

→ Probability Mass function (For discrete random var only)

Let X be a discrete random variable which assumes the values $x_0, x_1, \dots, x_n$. Let $P(X=x_i) = P(x_i) = f_i$ so value of f_i depends on x_i that's i . This func f_i is called probability mass function of random variable X . A particular value of f_i is called as probability mass.

• $f_i \geq 0$ and $\sum f_i = 1$

Q If $f(x) = kx$, $x = 1, 2, \dots, n$ be pmf of x . $k = ?$

$$\Rightarrow \sum_{x=1}^n kx = 1$$

$$\Rightarrow k = \frac{2}{n(n+1)}$$



Q Find PMF of no of heads when fair coin is tossed repeatedly until first tail appears

$$X=0 \quad P(X) = \frac{1}{2}$$

$$X=1 \quad P(X) = \frac{1}{2} \times \frac{1}{2}$$

$$X=2 \quad P(X) = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$$

X	0	1	2	3	...	n
P(X)	1/2	1/2 ²	1/2 ³	1/2 ⁴	...	(1/2) ⁿ

$$X=N \quad P(X) = \left(\frac{1}{2}\right)^{n+1}$$

→ Distribution function or cumulative distribution function

The distribution function of a random variable X (discrete/continuous) is given by

$$F(x) = P(-\infty < X \leq x) \quad -\infty < x < \infty$$

Thus if $x_i \leq x < x_{i+1}$ then

$$F(x) = P(X=x_0) + P(X=x_1) + P(X=x_2) + \dots + P(X=x_i)$$

Let's take ex

X	1	2	3	4	5	6
f _i	1/6	1/6	1/6	1/6	1/6	1/6

Now if $x < 1 \quad F(x) = P(X \leq x) = 0$

if $1 \leq x < 2 \quad F(x) = P(X \leq x) = f_1 = 1/6$

$2 \leq x < 3 \quad F(x) = P(X \leq x) = f_1 + f_2 = 2/6 = 1/3$

$3 \leq x < 4 \quad F(x) = P(X \leq x) = f_1 + f_2 + f_3 = 3/6 = 1/2$

$4 \leq x < 5 \quad F(x) = P(X \leq x) = f_1 + f_2 + f_3 + f_4 = 2/3$

$5 \leq x < 6 \quad F(x) = P(X \leq x) = f_1 + f_2 + f_3 + f_4 + f_5 = 5/6$

$6 \leq x < \infty \quad F(x) = \sum_{i=1}^{\infty} f_i = 1$



→ Properties of distribution function

- $F(x)$ is a monotonic non decreasing function
- $F(-\infty) = 0$ & $F(\infty) = 1$ hence $0 \leq F(x) \leq 1$
- $F(x)$ is continuous on right at all points & has jump discontinuity on left at $x = a$, the height of jump being equal to $P(X=a)$ i.e. $\lim_{x \rightarrow a^+} F(x) = F(a)$ and $F(a) - \lim_{x \rightarrow a^-} F(x) = P(X=a)$

- Suppose a & b are real nos such that $a < b$
then $P(a < X \leq b) = F(b) - F(a)$

$$P(a < X < b) = F(b) - F(a) - P(X=b)$$

$$P(a \leq X < b) = F(b) - F(a) - P(X=b) + P(X=a)$$

→ expectation / Mean of a discrete random variable

Let X be a discrete random variable whose dist. is

x	x_0	x_1	x_2	$\dots$	x_n
f_i	f_0	f_1	f_2	$\dots$	f_n

Then mean / expected value of X denoted by $E(X)$ or $m(X)$ or μ is

$$E(X) = x_0 f_0 + x_1 f_1 + \dots + x_n f_n$$

$E(X) = \sum f_i x_i$, provided series is absolutely convergent if above sum is ∞ series

→ Properties of $E(X)$

- $E(a) = a$ where $a \rightarrow$ constant
- $E(ax+b) = aE(X) + b$
- $E[ag(x)+b] = aE[g(x)] + E(b)$



→ Variance & standard deviation of X

$$\text{Var}(X) = E[(X - \mu)^2] \text{ where } \mu = E(X)$$

$$\sigma_X = +\sqrt{\text{Var}(X)}$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

$$\text{Var}(K) = 0 \quad K \rightarrow \text{constant}$$

Q

X	0	1	2	3	4	5	6	7
$P(X)$	0	K	$2K$	$2K$	$3K$	K^2	$2K^2$	$7K^2 + K$

$$K = ? \quad P(X < 6) = ? \quad \text{and} \quad P(0 < X < 5)$$

Find min value of x so that $P(X \leq x) > 1/2$

$$\leq P(X) = 1 \Rightarrow 10K^2 + 9K = 1$$

$$\Rightarrow 9K^2 + 10K^2 = 1$$

$$\Rightarrow 10K^2 + 10K - K - 1 = 0$$

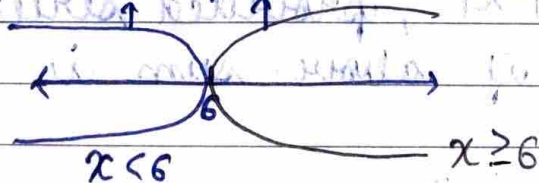
$$\Rightarrow 10K(K+1) - (K+1) = 0$$

$$\Rightarrow K = 1/10$$

X	0	1	2	3	4	5	6	7
$P(X)$	0	$1/10$	$2/10$	$2/10$	$3/10$	$1/100$	$2/100$	$17/100$

complement to each other.

Now,



$$P(X < 6) = 1 - P(X \geq 6)$$

$$= 1 - P(X=6) - P(X=7)$$

$$= 1 - \frac{2}{100} - \frac{17}{100} = \frac{81}{100} = 0.81$$



$$\begin{aligned}
 P(0 < X < 5) &= P(X=1) + P(X=2) + P(X=3) + P(X=4) \\
 &= 0.1 + 0.2 + 0.2 + 0.3 \\
 &= 0.80
 \end{aligned}$$

$$\begin{aligned}
 P(3 \leq X < 6) &= P(3) + P(4) + P(5) \\
 &= 0.2 + 0.3 + 0.01 \\
 &= 0.51
 \end{aligned}$$

Now, $P(X \leq x) > 1/2$

$$P(X \leq 1) = K = 0.1$$

$$P(X \leq 2) = 0.1 + 0.2 = 0.3$$

$$P(X \leq 3) = 0.1 + 0.2 + 0.2 = 0.5 = 1/2$$

$$P(X \leq 4) = 0.8 > 1/2$$

$$P(X \leq 5) = 0.81$$

$$P(X \leq 6) = 0.83$$

$$P(X \leq 7) = 1$$

$\Rightarrow$ min value of $x = 4$

Now,

$$\begin{aligned}
 F(x) &= \begin{cases} 0 & -\infty < x \leq 0 \\ 0 & 0 \leq x < 1 \\ 0.1 & 1 \leq x < 2 \\ 0.3 & 2 \leq x < 3 \\ 0.5 & 3 \leq x < 4 \\ 0.8 & 4 \leq x < 5 \\ 0.81 & 5 \leq x < 6 \\ 0.83 & 6 \leq x < 7 \\ 1 & x \geq 7 \end{cases} \\
 \downarrow \\
 \text{Distribution} & \\
 \text{Function} &
 \end{aligned}$$

x	1	2	3	4	5	6	7
$P(X=x)$	0.1	0.2	0.2	0.3	0.01	0.01	0.01

Q Let $F(x) = \begin{cases} 0 & -\infty < x < 0 \\ 0.2 & 0 \leq x < 1 \\ 0.6 & 1 \leq x < 3 \\ 1 & 3 \leq x < \infty \end{cases}$

Show that $F(x)$ is possible distribution func
 Determine spectrum and probability mass
 Hence find μ and σ

- i) $F(x)$ is monotonic and non decreasing
 and cont on right of all pts (and
 $F(-\infty) = 0$ & $F(\infty) = 1$
 So $F(x)$ is possible distribution function

Now, spectrum me us values $\leq$ ka sign hai

$\Rightarrow$ Spectrum = $\{0, 1, 3\}$

Also, $F(a) - \lim_{x \rightarrow a^-} F(x) = P(X=a)$

$\Rightarrow F(0) - \lim_{x \rightarrow 0^-} F(x) = P(0)$

$\Rightarrow P(0) = 0.2 - 0 = 0.2$

$\Rightarrow F(1) - \lim_{x \rightarrow 1^-} F(x) = P(1)$

$\Rightarrow P(1) = 0.6 - 0.2 = 0.4$

$\Rightarrow P(3) = 1 - 0.6 = 0.4$

x	0	1	3
$P(x)$	0.2	0.4	0.4



$$E(X) = (0 \times 0.2) + (1 \times 0.4) + (3 \times 0.4)$$

$$= 0 + 0.4 + 1.2$$

$$\mu = 1.6$$

$$E(X^2) = (0^2 \times 0.2) + (1^2 \times 0.4) + (3^2 \times 0.4)$$

$$= 0 + 0.4 + 3.6$$

$$= 4.0$$

$$\sigma = \sqrt{E(X^2) - [E(X)]^2} = \sqrt{4.0 - 2.56} = \sqrt{1.44}$$

$$\sigma = 1.2$$

$$Q \quad F(x) = \begin{cases} A & -\infty < x < -1 \\ B & -1 \leq x < 0 \\ C & 0 \leq x < 2 \\ D & 2 \leq x < \infty \end{cases} \quad \begin{aligned} F(-\infty) &= A = 0 \\ F(\infty) &= D = 1 \end{aligned}$$

$$P(X=0) = 1/6 \text{ and } P(X > 1) = 2/3, \quad A, B, C, D = ?$$

$$F(a) = \lim_{x \rightarrow a^-} F(x) = P(a)$$

$$B - 0 = P(X = -1)$$

$$B = P(X = -1)$$

$$C - B = 1/6$$

$$6C - 6B = 1 \quad (1)$$

$$P(X > 1) = 1 - P(X \leq 1)$$

$$P(X \leq 1) = 2/3$$

$$P(X \leq 1) = 1/3 = C$$

$$\Rightarrow B = \frac{1}{3} - \frac{1}{6} = \frac{1}{6}$$



→ Bernoulli Distribution $X \sim \text{Bernoulli}(p)$

A discrete random variable X is said to have a Bernoulli distribution if its PMF gets form

$$f(x) = \begin{cases} 1-p & x=0 \\ p & x=1 \\ 0 & \text{otherwise} \end{cases}$$

x	0	1
$P(x)$	$1-p$	p

where $p \in (0,1)$ is a parameter. Such a random variable can take only 2 values 0 & 1

We write $X \sim \text{Bernoulli}(p)$ and we say X is a Bernoulli variate

→ Bernoulli Trials (Finite and infinite)

If a random experiment has only 2 possible outcomes (0 & 1) and it's repeated independently then resulting sequence of trials is called a Bernoulli sequence of trials

Q If p is probability of success in a single trial find probability of $r (\leq n)$ successes in a Bernoulli seq of n trials

In a seq of n trials with outcomes failure (0) and success (1). Exactly ' r ' successes are to occur in nC_r ways.

Each such seq has r success & $n-r$ failures since trials are independent. The probability of each such seq is

$$[P(\text{success})]^r [P(\text{failure})]^{n-r}$$

so req probability is

$${}^nC_r [P(S)]^r [P(F)]^{n-r}$$



$$= {}^n C_r p^r (1-p)^{n-r} \rightarrow \text{Binomial law}$$

Q If 20 dates are named at random, what's the probability that 3 of them will be Sundays

$$n = 20, r = 3$$

Success = Sunday hona chahiye

$$P(\text{Sunday}) = \frac{1}{7}$$

$$\Rightarrow p = 1/7$$

$$\Rightarrow P(3 \text{ Sundays}) = {}^{20}C_3 \left(\frac{1}{7}\right)^3 \left(\frac{6}{7}\right)^{17}$$

or

Let E be the random experiment of naming a day against a date. Then this is a Bernoulli's trial E_{20} . Let in a trial, success denotes 'the named date is Sunday' & failure denotes 'the named date isn't Sunday'.

$$\text{In single trial} \rightarrow P(\text{Success}) = \frac{1}{7}$$

$$\Rightarrow P(\text{Failure}) = 1 - \frac{1}{7} = \frac{6}{7}$$

Now, for 3 Sundays, we need 3 trials

$$\Rightarrow P(3 \text{ Sundays}) = {}^{20}C_3 \left(\frac{1}{7}\right)^3 \times \left(\frac{6}{7}\right)^{17}$$

Q A & B play a game in which their chances to win are in ratio 3:2. Find A's chances of winning in 3 out of 5 games.

Let E be event of playing a game. This is a Bernoulli trial E_5 . Let success denote A's win &



A failure denotes A's loose

$$P(\text{success}) = \frac{3}{5}$$

$$P(\text{failure}) = 1 - \frac{3}{5} = \frac{2}{5}$$

For A to win 3 games, we need 3 trials

$$P(3 \text{ wins}) = {}^5C_3 \left(\frac{3}{5}\right)^3 \left(\frac{2}{5}\right)^2$$

$$p \rightarrow P(\text{success})$$

→ Binomial distribution $n \rightarrow$ no of trials

A discrete random variable is said to have a binomial distribution with parameters p ($0 < p < 1$) and n (a +ve integer) if its distribution is given by :-

spectra $\leftarrow x$ 0 1 2 ... n

PMF $\leftarrow f_i$ f_0 f_1 ... f_n

where PMF $f_i = {}^nC_i (p)^i (1-p)^{n-i}$

$$X \sim B(n, p)$$

↳ It represents x is a binomial dist.

Q Efficiency of fighter plane is that a probability of bomb hitting target is $2/5$. Plane is req to destroy a camp of enemy side & atleast 4 bombs are req for it. The plane carries 6 bombs. P that enemy camp is destroyed?

$$P(\text{success}) = 2/5$$

$$P(\text{failure}) = 1 - 2/5 = 3/5$$

where dropping a bomb is an event and success is hitting the target & failure is not hitting the target



Now, this event has 6 trials because plane has 6 bombs,

$$\Rightarrow X \sim B(6, p(s))$$

Case I: 4 bombs hit

$$\begin{aligned} f_4 &= {}^6C_4 \left(\frac{2}{5}\right)^4 \left(\frac{3}{5}\right)^2 \\ &= \frac{6 \times 5}{2} \times \frac{16}{5^6} \times 9 = \frac{48 \times 9}{5^5} = \frac{432}{3125} \\ &= \frac{2160}{15625} \end{aligned}$$

Case II: 5 bombs hit

$$\begin{aligned} f_5 &= {}^6C_5 \left(\frac{2}{5}\right)^5 \left(\frac{3}{5}\right) \\ &= 6 \times \frac{32 \times 3}{15625} = \frac{576}{15625} \end{aligned}$$

Case III: 6 bombs hit

$$f_6 = {}^6C_6 \left(\frac{2}{5}\right)^6 = \frac{64}{15625}$$

$$\begin{aligned} \Rightarrow P(\text{enemy camp dest}) &= C_1 + C_2 + C_3 \\ &= \frac{2160 + 576 + 64}{15625} \\ &= \frac{2790}{15625} = \frac{558}{3125} \end{aligned}$$



→ Theorem

If X has a binomial distribution with the factors n & p

i) mean = np

ii) variance = npq
where $q = 1-p$

Let's consider a $X \sim B(n, p)$

$$P_0 = {}^nC_0 p^0 (1-p)^n$$

$$P_1 = {}^nC_1 p (1-p)^{n-1}$$

⋮

$$P_n = {}^nC_n p^n (1-p)^0$$

X	0	1	⋮	n
P_i	${}^nC_0 (1-p)^n$	${}^nC_1 p (1-p)^{n-1}$	⋮	${}^nC_n p^n$

i) Mean = $E(X) = \sum X_i P_i$

$$= 0 \cdot {}^nC_0 (1-p)^n + 1 \cdot {}^nC_1 p (1-p)^{n-1} + 2 \cdot {}^nC_2 p^2 (1-p)^{n-2} + \dots + n \cdot {}^nC_n p^n$$

$$= \sum_{i=0}^n i \cdot {}^nC_i p^i (1-p)^{n-i}$$

$$= \sum_{i=0}^n \frac{i \cdot n!}{i! (n-i)!} (p^{i-1} \cdot p) (1-p)^{n-i}$$

$$= \sum_{i=0}^n \frac{n [n-1]!}{(i-1)! (n-i)!} (p^{i-1} \cdot p) (1-p)^{n-i}$$



$$\begin{aligned}
 &= \sum_{i=0}^n n \cdot n^{-1} C_{i-1} p^{i-1} \cdot p (1-p)^{n-i} \\
 &= np \sum_{i=0}^n n^{-1} C_{i-1} p^{i-1} (1-p)^{n-i} \\
 &= np (p + (1-p))^{n-1} = np (1)^{n-1} \\
 &= np
 \end{aligned}$$

Q Mean & S.D of binomial dist are 4 and $\sqrt{8/3}$.
Find n & p , evaluate $P(X=0)$

$$np = 4$$

$$np(1-p) = 8/3$$

$$\frac{1}{1-p} = \frac{4}{8} \times 3 = \frac{3}{2}$$

$$2 = 3 - 3p$$

$$p = \frac{1}{3}$$

$$1-p = \frac{2}{3}$$

$$\Rightarrow p = 12$$

$$P(X=0) = {}^n C_0 \left(\frac{2}{3}\right)^n = {}^{12} C_0 \left(\frac{2}{3}\right)^{12} = \left(\frac{2}{3}\right)^{12}$$

Q Mean = 3, var = 3/2 $P(\text{atmost 3 success}) = ?$

$$np = 3$$

$$npq = 3/2$$

$$\frac{1}{q} = 2$$

$$\Rightarrow 1-p = \frac{1}{2} \Rightarrow p = \frac{1}{2}$$

$$\Rightarrow n = 6$$

$$P(0 \text{ succ}) = {}^6 C_0 (1/2)^6 = 1/64$$

$$P(1 \text{ succ}) = {}^6 C_1 (1/2)^6 = 6/64$$

$$P(2 \text{ S}) = {}^6 C_2 (1/2)^6 = 15/64$$

$$P(3 \text{ S}) = {}^6 C_3 (1/2)^6 = 20/64$$

$$P(\text{atmost 3S}) = 42/64 = 21/32$$



- Q P (pen manufactured by company being defective) = $1/10$. 12 pens are made
- i) P (exactly 2 are defective)
 - ii) P (none is defective)
 - iii) P (atmost 2 are defective)

Let E is the event to make a pen and a defective pen is success and non defective pen is failure. $\therefore X \sim B(12, 1/10)$

gn,

1 trial $P(\text{success}) = 1/10$

& $P(\text{failure}) = 9/10$

- i) Exactly 2 defective

$$= {}^{12}C_2 \left(\frac{1}{10}\right)^2 \left(\frac{9}{10}\right)^{10} = 66 \times \frac{9^{10}}{10^{12}}$$

- ii) None is defective

$$= {}^{12}C_0 \left(\frac{9}{10}\right)^{12} = \left(\frac{9}{10}\right)^{12}$$

- iii) Atmost 2 defective

$$P(X=0) = \left(\frac{9}{10}\right)^{12}$$

$$P(X=1) = {}^{12}C_1 \left(\frac{1}{10}\right) \left(\frac{9}{10}\right)^{11} = 12 \times \frac{9^{11}}{10^{12}}$$

$$P(X=2) = 66 \times \frac{9^{10}}{10^{12}}$$

$$= \left(\frac{9}{10}\right)^{12} + \frac{12 \times 9^{11}}{10^{12}} + \frac{66 \times 9^{10}}{10^{12}}$$

Boy $\rightarrow$ success

Page

Date



- Q A family has 6 children. Find probability that
i) 3B & 3G ii) fewer boys than girls

$$i) {}^6C_3 \left(\frac{1}{2}\right)^6 = \frac{20}{64}$$

$$ii) {}^6C_0 \left(\frac{1}{2}\right)^6 + {}^6C_1 \left(\frac{1}{2}\right)^6 + {}^6C_2 \left(\frac{1}{2}\right)^6 \\ = \left(\frac{1}{2}\right)^6 \times (1 + 6 + 15) = \frac{22}{64}$$

$\rightarrow$ Poisson distribution $\rightarrow P(X=i) = \frac{1}{i!} (\mu)^i e^{-\mu}$

A discrete random variable X is said to have a poisson distribution with parameter $\mu > 0$ if its dist. is given by

X	0	1	...	n
f_i	f_0	f_1	...	f_n

where PMF, $f_i = P(X=i) = \frac{e^{-\mu} \mu^i}{i!}$

$$X \sim \text{poisson}(\mu)$$

$\rightarrow$ Theorem

If $X \sim \text{poisson}(\mu)$

• Mean = Variance = μ

Proof :-

$$P(X=i) = \frac{1}{i!} \left(\frac{\mu^i}{e}\right) e^{-\mu}$$



Mean = $E(X) = \sum_{i=0}^{\infty} i \cdot P(X=i)$ if $P(X=i) = \frac{e^{-\mu} \mu^i}{i!}$

$$= \sum_{i=0}^{\infty} i \frac{e^{-\mu} \mu^i}{i!} = e^{-\mu} \sum_{i=0}^{\infty} \frac{\mu^i}{(i-1)!}$$

$$= e^{-\mu} \left[\frac{\mu}{0!} + \frac{\mu^2}{1!} + \frac{\mu^3}{2!} + \dots + \frac{\mu^{\infty}}{(i-1)!} \right]$$

$$= e^{-\mu} \left[\mu + \mu^2 + \frac{\mu^3}{2} + \frac{\mu^4}{6} + \dots + \frac{\mu^{\infty}}{(i-1)!} \right]$$

$$= \mu e^{-\mu} \left[1 + \mu + \frac{\mu^2}{2!} + \frac{\mu^3}{3!} + \dots + \frac{\mu^{\infty}}{(i-1)!} \right]$$

expression for e^{μ}

$$= \mu e^{-\mu} \times e^{\mu} = \mu$$

$$= \mu$$

(if) poisson $\sim X$

(if) poisson $\sim X$ if

$\mu = \text{mean}$

$$\mu = \sum_{i=0}^{\infty} i \cdot P(X=i)$$



→ Poisson Approximation to Binomial Distribution
In a binomial dist when n becomes very large p is small enough that $np \approx \text{constant}$ & becomes equal to poisson variate

Let X follows binomial dist with the PMF
 $f_i = P(X=i) = {}^n C_i p^i (1-p)^{n-i}$

If $n \rightarrow \infty$ and $p \rightarrow 0$ such that $np = \mu$ then

$$\lim_{n \rightarrow \infty} {}^n C_i p^i (1-p)^{n-i} = \frac{e^{-\mu} \mu^i}{i!}$$

Q Let box has 200 fuses. Experience tells 2% of such fuses are defective. Let's consider the exp of drawing fuse whether this is defective or not

$$n = 200$$

$$n \gg p$$

$$p = \frac{2}{100}$$

$$\Rightarrow np = \mu = 4$$

$$\Rightarrow P(X=i) = \frac{e^{-4} (4)^i}{i!}$$

Q A car firm has 2 cars which it hires out day by day. The no of demands on a car on each day is distributed as a poisson dist as avg no of demands per day is 1.5. Calculate proportion of days in which neither car is used & proportion of days on which some demand is refused

↳ both cars are used

$$e^{-1.5} = 0.2231$$



X is a discrete random variable denoting no. of demands for a car made each day and $\mu = 1.5$

- i) Prop on which neither car is used
 $\Rightarrow$ There is no demand for car

$$\Rightarrow X = 0$$

$$\Rightarrow P(X=0) = \frac{e^{-1.5} \mu^0}{0!} = 0.2231$$

- ii) $\Rightarrow$ Both cars used
 hence $X > 2$

Also,

$$P(X > 2) = 1 - P(X \leq 2)$$

$$\begin{aligned} P(X \leq 2) &= P(X=0) + P(X=1) + P(X=2) \\ &= 0.2231 + 0.5(0.2231) + \frac{0.5}{2}(0.2231) \end{aligned}$$

$$= 0.2231 \left(\frac{4.5}{2} \right)$$

$$= 0.501975$$

$$\Rightarrow P(X > 2) = 0.498025$$

- Q In a certain factory turning razor blades, there is a small chance $1/500$ for any blade to be defective. Blades are in packets of 10. Use poisson dist. to calculate approx no. of packets having i) no defective, ii) 1 defective & iii) 2 defective blades in a consignment of 10,000 packets

$$[e^{-0.02} = 0.9802]$$



An avg, 1 blade of 500 is defective, so avg defective blades in packet of 10 is $10 \times \frac{1}{500}$

$$= 0.02 \Rightarrow \mu = 0.02$$

$X \sim \text{poission}(0.02)$

$$P(X=i) = \frac{e^{-\mu} \mu^i}{i!}$$

$$i) P(X=0) = e^{-0.02}$$

$$\text{No of packets} = 10,000 \times e^{-0.02} \\ = 9802$$

$$ii) P(X=1) = 0.02 e^{-0.02}$$

$$\text{No of packet} = 10,000 \times 0.02 \times 0.9802 \\ = 196.04 \\ \approx 196$$

$$iii) P(X=2) = \frac{0.02}{2} e^{-0.02}$$

$$\text{No of packets} = 198.02$$

$$\approx 98 \times 2 = 196$$